

PASCO TRI-CITIES, WA, USA

WMO: 727845

Lat: 46.267N

Lon: 119.117W

Elev: 124

StdP: 99.84

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24163

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-11.6	-8.3	-16.0	0.9	-9.8	-12.4	1.3	-6.5	12.8	9.1	11.5	8.1	1.8	330	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	37.3	20.6	35.7	19.8	33.7	19.1	21.9	34.1	20.8	33.1	19.9	31.6	3.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	26.2	16.6	12.0	24.9	15.4	11.1	23.9	64.2	34.2	60.3	33.1	56.9	31.5	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	8.0	DB	-15.9	40.7	5.2	1.8	-19.7	42.0	-22.7	43.0	-25.6	44.0	-29.4	45.3	(4)
(5)			WB	-16.3	24.2	5.1	2.0	-20.0	25.6	-23.0	26.8	-25.9	27.9	-29.6	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	1.7	3.7	7.8	11.7	16.3	20.1	24.1	23.1	18.0	11.3	5.2	1.1	(6)	
	DBStd	8.83	4.93	4.37	3.76	3.16	3.77	3.30	3.07	2.98	3.31	3.83	4.84	4.84	(7)	
	HDD10.0	1006	259	179	84	17	2	0	0	0	1	29	156	278	(8)	
	HDD18.3	2751	516	411	326	199	85	18	1	2	44	219	394	535	(9)	
	CDD10.0	1758	1	3	17	69	198	302	438	408	241	69	11	2	(10)	
	CDD18.3	461	0	0	0	1	23	69	181	151	35	1	0	0	(11)	
	CDH23.3	7222	0	0	2	53	459	1089	2746	2187	656	30	0	0	(12)	
	CDH26.7	3627	0	0	0	9	168	477	1564	1159	246	4	0	0	(13)	
(14) Wind	WSAvg	2.9	2.7	2.9	3.4	3.5	3.3	3.4	2.8	2.6	2.5	2.6	2.7	2.6	(14)	
(15) Precipitation	PrecAvg	197	27	18	17	14	16	10	5	9	8	14	26	29	(15)	
	PrecMax	347	98	46	44	47	58	31	38	42	41	48	74	88	(16)	
	PrecMin	96	5	0	0	0	0	0	0	0	0	0	2	4	(17)	
	PrecStd	53	18	12	11	11	12	8	7	11	9	12	17	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	16.9	22.3	27.8	33.9	37.3	40.5	38.6	34.1	27.0	19.9	15.4	(19)	
		MCWB	10.2	11.0	12.7	15.9	18.5	21.1	21.7	20.3	19.7	16.1	14.4	10.8	(20)	
	2%	DB	12.4	14.1	19.0	24.5	31.1	34.1	37.9	36.7	31.9	23.0	17.1	12.6	(21)	
		MCWB	8.3	9.0	11.0	14.3	17.7	19.3	20.7	20.3	18.3	14.6	11.4	8.8	(22)	
	5%	DB	10.4	12.4	17.1	22.1	28.1	31.7	36.4	34.8	29.6	21.0	14.5	10.2	(23)	
		MCWB	7.1	7.9	10.2	12.9	16.2	18.2	20.2	19.4	17.4	13.2	10.0	7.1	(24)	
	10%	DB	8.4	10.8	15.2	19.5	25.8	29.1	34.5	32.8	27.3	18.8	12.5	8.0	(25)	
		MCWB	5.9	6.8	8.9	11.1	15.1	16.9	19.4	18.7	16.5	12.1	8.7	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	11.7	13.7	17.3	21.1	22.3	24.9	22.7	20.6	17.1	15.2	11.2	(27)	
		MCDB	14.1	16.0	20.0	26.2	29.6	34.3	36.2	34.8	31.6	24.4	19.2	14.9	(28)	
	2%	WB	8.7	9.9	11.9	15.2	18.7	20.4	22.4	21.4	19.2	15.4	12.0	9.0	(29)	
		MCDB	11.8	13.5	17.4	23.1	28.1	32.2	34.8	33.9	29.6	22.0	15.7	12.1	(30)	
	5%	WB	7.4	8.2	10.7	13.5	17.2	19.0	21.1	20.4	18.1	14.1	10.6	7.5	(31)	
		MCDB	10.1	11.8	15.7	20.8	26.7	29.7	33.8	32.5	27.6	19.4	13.6	10.0	(32)	
	10%	WB	5.9	7.1	9.7	11.9	15.8	17.7	20.1	19.4	17.0	12.8	9.3	5.7	(33)	
		MCDB	8.3	10.1	14.6	18.4	23.9	27.5	32.4	30.7	26.0	17.7	12.2	7.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	10.3	13.1	14.5	15.5	15.8	18.6	18.2	17.4	13.8	9.5	6.9	(35)	
		MCDBR	10.0	12.1	16.0	19.0	20.3	20.4	21.9	21.3	21.0	16.2	11.4	9.6	(36)	
	5% WB	MCWBR	6.9	7.8	9.1	9.7	8.5	7.7	7.6	7.5	8.7	8.5	7.1	6.7	(37)	
		MCWBR	9.7	11.3	14.5	18.3	17.9	19.2	20.8	20.1	19.5	14.9	10.5	9.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.290	0.292	0.306	0.345	0.362	0.360	0.363	0.359	0.343	0.313	0.297	0.285	(40)	
		taud	2.557	2.569	2.546	2.403	2.378	2.405	2.400	2.405	2.440	2.531	2.540	2.547	(41)	
	Ebn at Noon		828	900	931	912	904	904	896	887	872	857	807	791	(42)	
		Edh at Noon	62	74	87	110	117	114	113	108	96	76	62	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.18	2.26	3.57	5.29	6.41	7.20	7.69	6.45	4.78	2.96	1.54	0.97	(44)	
	RadStd		0.15	0.29	0.25	0.31	0.40	0.35	0.25	0.35	0.27	0.22	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.67	N/A	N/A	N/A	+144	+70
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A

Nomenclature: See separate page